

Digital Heroes – Task A Submission Report

Website name =PagePulse – Website Intelligence

1. Project Overview

PagePulse – Website Intelligence is a responsive multi-page marketing website developed for a fictional B2B software company.

The purpose of the website is to present PagePulse as a website monitoring and website-quality platform that helps teams understand website performance, identify issues, and improve their overall web workflow.

The project was built from scratch using **HTML and CSS**, with a focus on responsive design, semantic structure, accessibility, SEO fundamentals, performance, and a consistent visual experience across all pages.

2. Pages Developed

The website consists of four connected pages:

- **Home page** : The Home page introduces PagePulse and communicates the main value of the product.

It includes:

- Brand introduction and navigation
- Main hero section
- Product/value proposition
- Key website-quality benefits
- Calls-to-action
- Supporting sections explaining the product

- **Product Page:** The Product page explains what PagePulse provides and how it can help teams manage website quality.

It focuses on:

- Product capabilities
- Website monitoring concepts
- Quality and performance insights
- Benefits for teams and website workflows
- Clear calls-to-action

- **Pricing Page:** The Pricing page presents the available PagePulse plans in a simple and easy-to-compare format.

It includes:

- Different pricing plans
 - Feature comparisons
 - Clear plan information
 - Calls-to-action
 - FAQ-style information for common questions
- **Contact Page:** The Contact page provides a direct way for visitors to communicate with the PagePulse team.

It includes:

- Contact introduction
- Email and response-time information
- Name, email, company, and message fields
- Accessible form labels
- Contact CTA
- Link to the Pricing/FAQ section

3. Technical Implementation

The website was developed with a simple and lightweight front-end structure.

Technologies used: HTML5 , CSS3

Implementation highlights:

- Semantic HTML elements such as header, nav, main, section, and footer
- Reusable CSS styling across all pages
- Responsive layouts for different screen sizes
- Consistent navigation between all four pages
- Proper heading hierarchy
- Form labels connected to their respective inputs
- Mobile-friendly viewport configuration
- Clean and organized page structure

The project avoids unnecessary frameworks and keeps the **implementation lightweight and easy to maintain.**

4. Responsive Design

Responsive behavior was considered throughout the website so that the pages remain usable on both desktop and mobile screens.

The layout, typography, spacing, navigation, content sections, and form elements were designed to adapt to smaller screen sizes.

The website was also **tested using a mobile Lighthouse configuration** to check its performance and usability under a mobile browsing environment.

5. Accessibility

Accessibility was considered while building the interface.

Key accessibility practices included:

- Semantic HTML structure
- Proper heading hierarchy
- Descriptive link text
- Form fields with associated labels
- Keyboard-focusable interactive elements
- Sufficient color contrast
- Mobile-friendly touch targets
- Valid language declaration
- Accessible viewport configuration

Lighthouse accessibility testing resulted in a 100/100 score.

6. SEO & HTML Quality

Basic SEO practices were implemented across the website.

These include:

- Page-specific <title> elements
- Meta descriptions
- Proper HTML document structure
- lang="en" declaration
- Descriptive navigation links
- Crawlable internal page links
- Structured content hierarchy
- Open Graph metadata where appropriate
- Structured data considerations

The website achieved a 100/100 Lighthouse SEO score during testing.

7. Performance Optimization & Testing

The website was tested using Google Lighthouse 13.4.0 with a mobile device emulation and slow 4G network conditions.

Lighthouse Results:

Category	Score
Performance	100/100
Accessibility	100/100
Best Practices	100/100
SEO	100/100

Core Performance Metrics:

Metric	Result
First Contentful Paint (FCP)	1.0 s
Largest Contentful Paint (LCP)	1.3 s

Total Blocking Time (TBT)	0 ms
Cumulative Layout Shift (CLS)	0
Speed Index	1.0 s

The Lighthouse test also reported **22 passed performance-related audits**, with no browser console errors detected during the test.

Some Lighthouse diagnostics suggested further optional optimization opportunities, such as CSS minification and caching configuration. These are primarily deployment-level optimizations and do not prevent the current implementation from achieving a 100-performance score in the test environment.

8. Cross-Page Navigation

All four pages are connected through a consistent navigation system:

Home → Product → Pricing → Contact

The navigation is kept consistent across the website so users can move between the main sections without confusion.

The Contact page also provides a route back to the Pricing/FAQ information for users who need quick answers before contacting the team.

9. AI Usage

AI tools were used during the development process for assistance with website planning, content structuring, code suggestions, debugging, reviewing implementation decisions, and improving the overall presentation.

The generated suggestions were reviewed and adapted during development rather than being used without verification. The final website was manually tested, including navigation, responsive behavior, accessibility checks, and Lighthouse testing.

10. Final Deliverables

The final submission includes:

- PagePulse multi-page marketing website
- Home, Product, Pricing, and Contact pages
- Responsive HTML/CSS implementation
- Lighthouse testing results
- GitHub repository [Khushi289/Digital-Heroes](#)
- Live website URL [PagePulse - Website Intelligence](#)

Project: PagePulse – Website Intelligence

Developed for: Digital Heroes – Task A